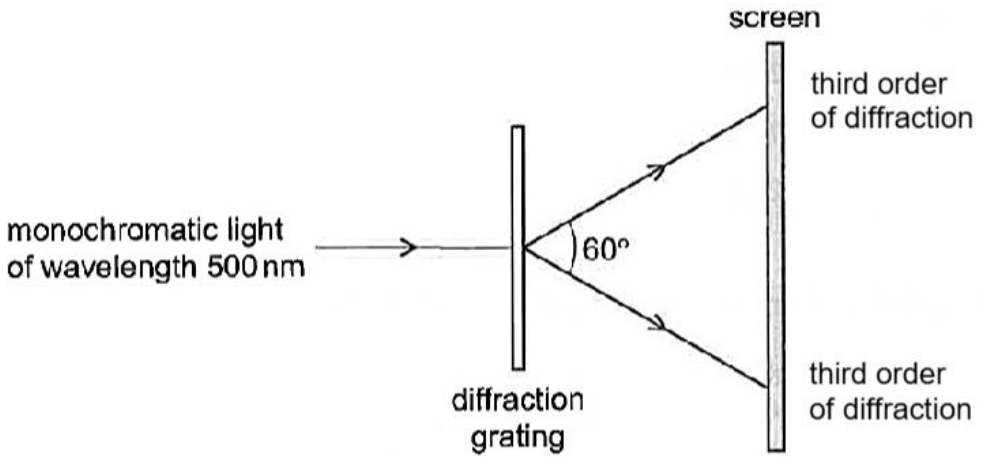
16	Monochromatic light of wavelength 500 nm is directed perpendicular onto a diffraction grating. The third order diffraction lines are observed at 60° to each other.



What is the number of diffraction lines per millimetre of the grating?

	A	333	**B**	507	**C**	667	**D**	333000

L2 **Answer: A**

$d \sin\theta = n\lambda$
$(1/N) \sin\theta = n\lambda$
$N = \sin\theta / n\lambda$
where
$\theta = 60°/2 = 30°$
$n = 3$
$\therefore N = \sin 30° / (3 \times 500 \times 10^{-9})$
$= 3.33 \times 10^{5}$ lines per metre
$= 3.33 \times 10^{2} = 333$ lines per millimetre